

Evolution of SoCs

It may be interesting to note that today, we have much more computing power in our phones than NASA had at the time of landing on the Moon. It is because of the single chip which functions as the brain and the heart of the cellphones and laptops. It has been a long time since the SoC revolution started. It began somewhere in the mid 1990s when semiconductor processing technology was reaching 350 and 250 nanometers, at which the major processing elements of complete system products could be placed on a single die for the first time.

The semiconductor industry traditionally lagged in the development of the fabric that ties the chip's individual components within the chip. SoC interconnect did not receive the type of research and development focus it required and suffered from routing congestion, physical timing closure problems, increased die size, market delays, efficiency losses, bottlenecks and frivolous performance potential. Recently the interconnect fabric technology has evolved to catch up with state of the art CPU and GPU technology.



This would cost you a fortune back in the day

After the synthesis of the first SoCs as we just saw, several other companies and developers started developing SoCs to suit the function of their device. The next major breakthrough came in the handheld gaming industry. SoCs were the perfect vehicle to carry this idea and were employed thoroughly throughout the 80s and the 90s in the various handheld gaming devices and even digital notepads (although their popularity never really took off). The biggest breakthrough in terms of commercial success came in the form of the device which we can't imagine living without now - the cellphone. SoCs made it possible to have mobile devices which were capable of making communication with other such devices - and eventually, do a lot more than that. SoCs now are way more sophisticated than they have ever been. But the 90s was a very important decade for the synthesis and implementation of SoCs.